INVESTIGATING THE RELATIONSHIP BETWEEN HOUSEHOLD INFLATION EXPECTATIONS AND NEWS CONTEXT USING SENTIMENT ANALYSIS

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Abstract

This study analyzes how news background shapes Russian households' median inflation expectations. First, under the rational inattention hypothesis, expectations were modeled from past CPI values via a Kalman filter, revealing a systematic bias relative to FOM survey data, implying use of additional information sources. Second, the effect of news on the gap between model- and survey-based expectations was examined using created sentiment-weighted indices for six economic and political topics. Households with savings responded to a wider range of macroeconomic news, while those without reacted mainly to fiscal news, highlighting the need to account for heterogeneity in expectation analysis.

Introduction

Inflation expectations are a crucial benchmark for the Bank of Russia's monetary policy. Empirical evidence shows that households, including those in Russia, systematically overestimate future inflation, possibly due to incomplete information. The concept of rational inattention explains this by suggesting attention focuses on the most salient signals, often originating from news [7, 9]. The media's influence has been formalised in epidemiological models [1], with modifications indicating expectations may also be shaped by non-inflationary news, where positive and negative sentiments have differing effects [2, 3, 6].

Previous Russian studies applying rational inattention focused on negative news as costlier to ignore [9] but did not model expectation formation within this framework. Models with rational inattention have been successfully applied to U.S. household data [5, 8]. This study fills the gap by modeling expectations based on official inflation data accounting for rational inattention, then assessing how sentiment-weighted thematic news indices explain discrepancies between model-based and survey-based Russian household inflation expectations.

Construction of Time-Series News Indices

In the study an innovative methodology has been developed for constructing time-series news indices, integrating the aggregation of a large text corpus with thematic modeling and sentiment analysis. The methodology involves estimating the full probabilistic distribution of themes using additive regularization of thematic models on a corpus of hundreds of thousands of news items from Russian media for the period 2010–2024.

The construction of monthly news indices is carried out by calculating the frequency of themes as the sum of probabilities of a text belonging to a theme, divided by the total number of news items in the month, which accounts for the possibility of multiple themes being revealed in one text and avoids overestimating the importance of themes during periods with a high volume of news. Sentiment analysis is conducted using a BERT-based model that evaluates tonality, which is a key factor in the perception of inflation.

Key findings include the identification of several themes, such as financial market, legislation, foreign trade relations, fiscal policy, monetary policy, and diplomatic relations, explaining aspects of the news field influencing expectations, with interpretability through key words. The dynamics of the obtained indices correlate with real events: positive indices for fiscal policy rise during periods of import substitution and anti-crisis measures, negative ones for foreign trade relations spike during times of sanctions, while positive ones for the financial market are high during recovery periods and fall during ruble collapses.

Modelling the Formation of Household Inflation Expectations Using Bayesian Learning

The formation of Russian household inflation expectations was modelled considering information constraints, addressing noise in median survey expectations caused by respondent inattention and heterogeneity. The framework is a Bayesian learning model with rational inattention, positing limited cognitive resources but a rational response to information [8].

The agent minimises expected quadratic forecast error under an information capacity constraint formalised by Shannon's mutual information. Sims provided a numerical solution [7]; Maćkowiak and Matějka an analytical one [4]. This study employs Maćkowiak and Matějka's formulation describing expectation formation via a time-invariant state-space model:

$$\pi_t = \alpha + \varphi\pi_{t-1} + \eta_t,\ \eta_t \sim \mathcal{N}(0, \delta_\eta^2)$$
$$S_t = \pi_t + w_t,\ w_t \sim \mathcal{N}(0, \delta_w^2)$$

where π_t is monthly inflation (year-over-year), a stationary Gaussian AR(1) process unobserved by households; S_t is a noisy signal. Maćkowiak and Matějka show the optimal signal under rational inattention is a one-dimensional Gaussian estimate of current inflation.

Inflation expectations are modelled via a Kalman filter, enabling Bayesian updating with linear dynamics and Gaussian noise. The signal is the monthly Consumer Price Index (CPI) year-over-year, matching the 12-month horizon in FOM surveys. The process includes prior forecasting and posterior updating, incorporating new CPI data, with an endogenous Kalman gain reflecting trust in new information based on its precision and noise.

Results obtained

Based on the obtained news indices, an econometric model has been estimated to assess their impact on inflation expectations of Russian households, allowing a transition from describing the informational background to analyzing its real influence on economic behavior. This approach is particularly important for identifying mechanisms of deviation from rational expectations in conditions of noisy media signals, providing the Central Bank of Russia with data to optimize communications and stabilize expectations.

The methodology involves the use of multiple linear regression, where the dependent variable is the gap between survey-based inflation expectations and their model estimates (Kalman filter), and the explanatory variables are lagged news indices across six themes, separated into positive and negative sentiment.

Key findings show that households with savings, possessing a higher level of financial literacy and education, respond to a wide range of macroeconomic news, they consider signals from monetary policy and legislation of various sentiments, as well as financial market news. In contrast, households without savings, with a lower level of economic knowledge, primarily rely on fiscal news of a social nature: negative fiscal messages increase the gap, intensifying the overestimation of expectations due to everyday

consumer experience, while positive ones reduce it. The directions of influence from different sentiments align with the results of previous works, such as Lamla and Lein, where negative media signals about inflation worsen forecast accuracy, confirming the asymmetry of perception [2].

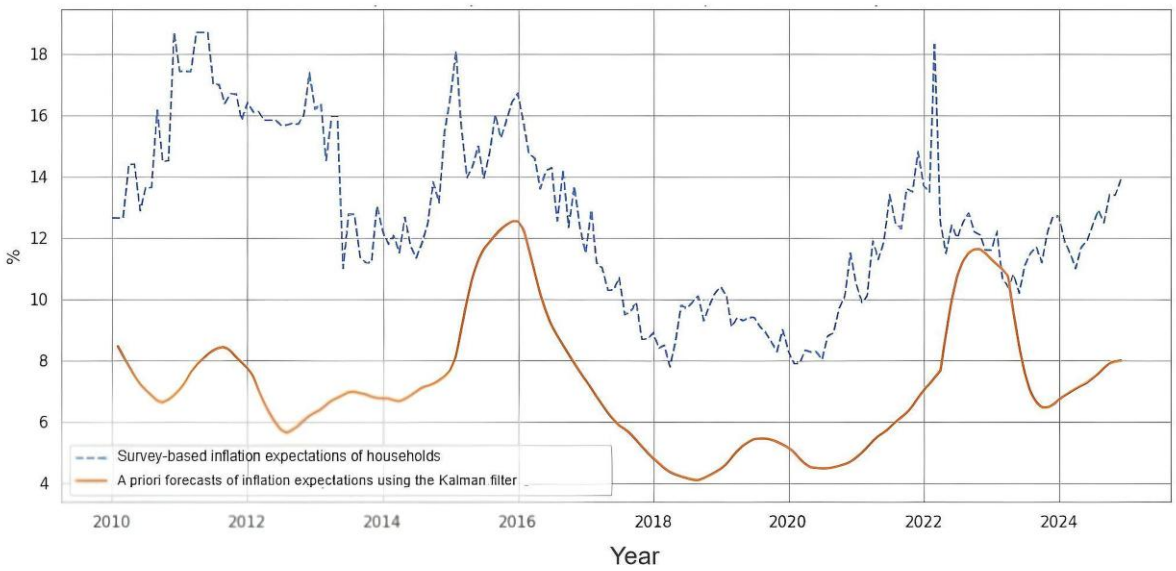

Figure 1. Results of Applying the Kalman Filter Based on the Consumer Price Index
Source: Authors' calculations

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